EXPERIMENT 24: UNIX System Calls for File Management

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/stat.h>

int main() {

  int fd;
  char buffer[100];

  // Create a file

  fd = creat("sample.txt", 0644);

  if (fd == -1) {
    perror("File creation failed");
    return 1;
  }

  printf("File created successfully.\n");

  close(fd);

  // Open file for writing

  fd = open("sample.txt", O_WRONLY | O_APPEND);
```

```
if (fd == -1) {
   perror("File opening failed");
   return 1;
}

printf("File opened for writing.\n");

write(fd, "Hello, World!\n", 14);

printf("Data written successfully.\n");

close(fd);

// Open file for reading

fd = open("sample.txt", O_RDONLY);

if (fd == -1) {
   perror("File opening failed");
   return 1;
}

printf("File opened for reading.\n");

int bytes = read(fd, buffer, sizeof(buffer) - 1);

if (bytes == -1) {
   perror("Read failed");
   close(fd);
   return 1;
}
```

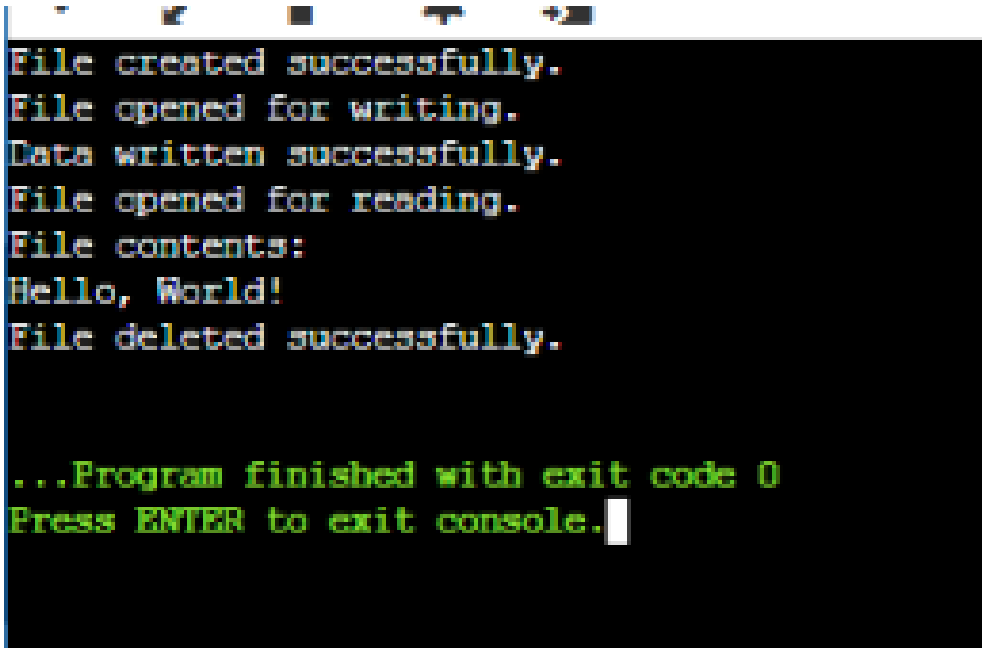
```
    buffer[bytes] = '\0';

    printf("File contents:\n%s", buffer);

    close(fd);

    // Delete file

    if (unlink("sample.txt") == 0)
        printf("File deleted successfully.\n");
    else
        perror("Delete failed");

    return 0;
}
```

Output: